

Air University



Network Security (Lab)

Year 2026

Mr. Haseeb Ahmed

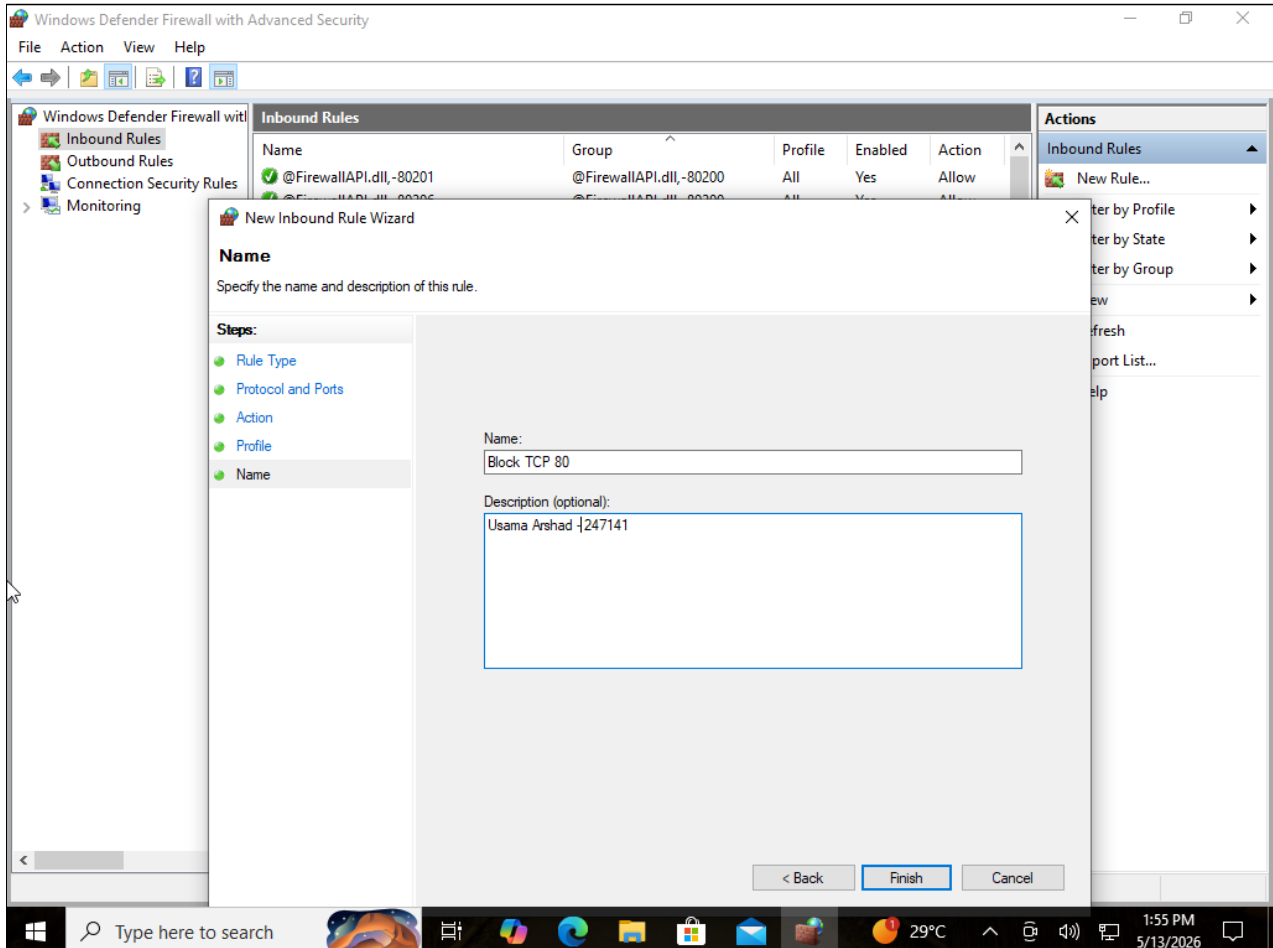
Academic Year: 2024 – 2028

Department of Cyber Security

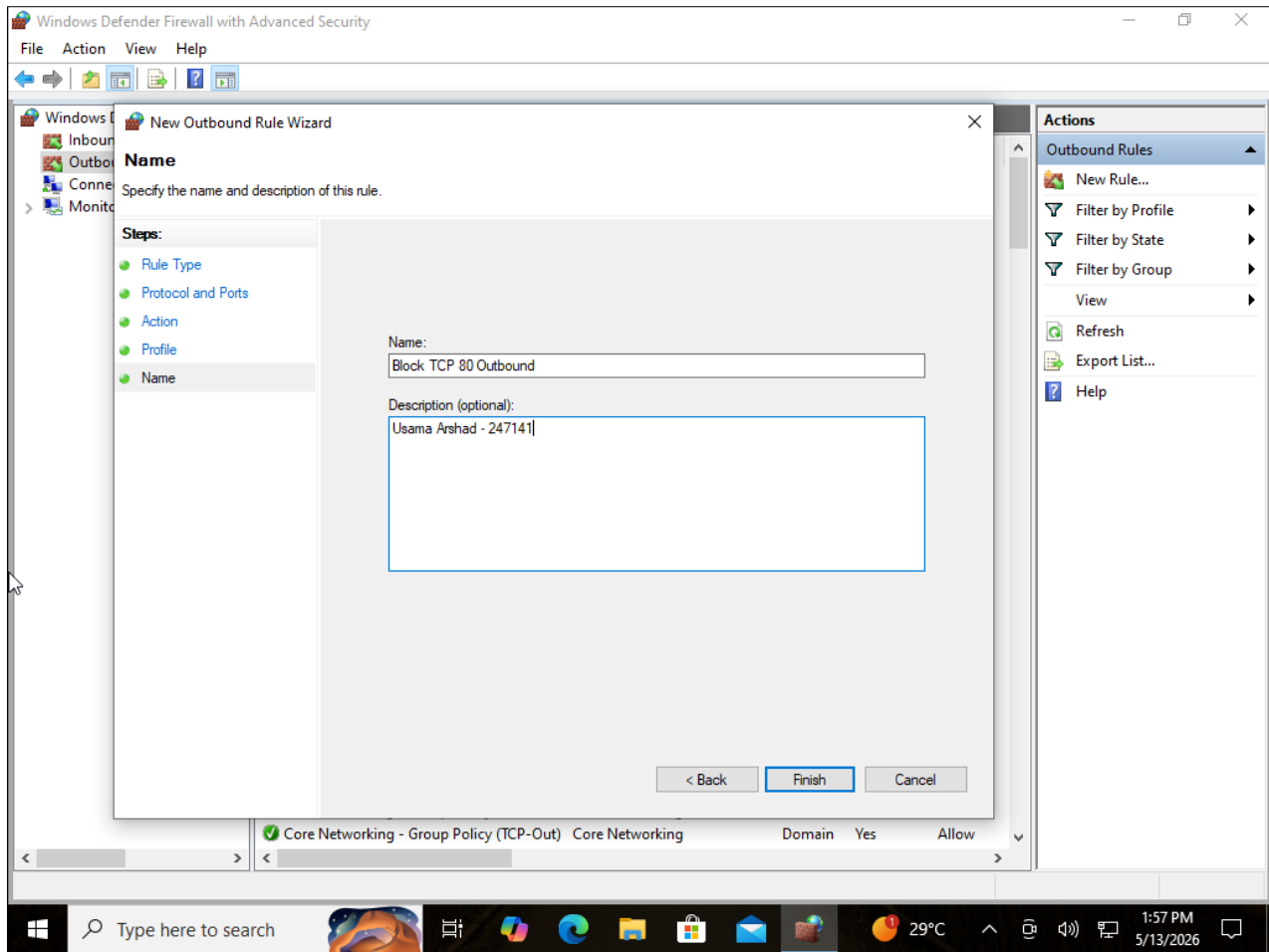
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BS – Cyber Security – Fall – 24 (A)

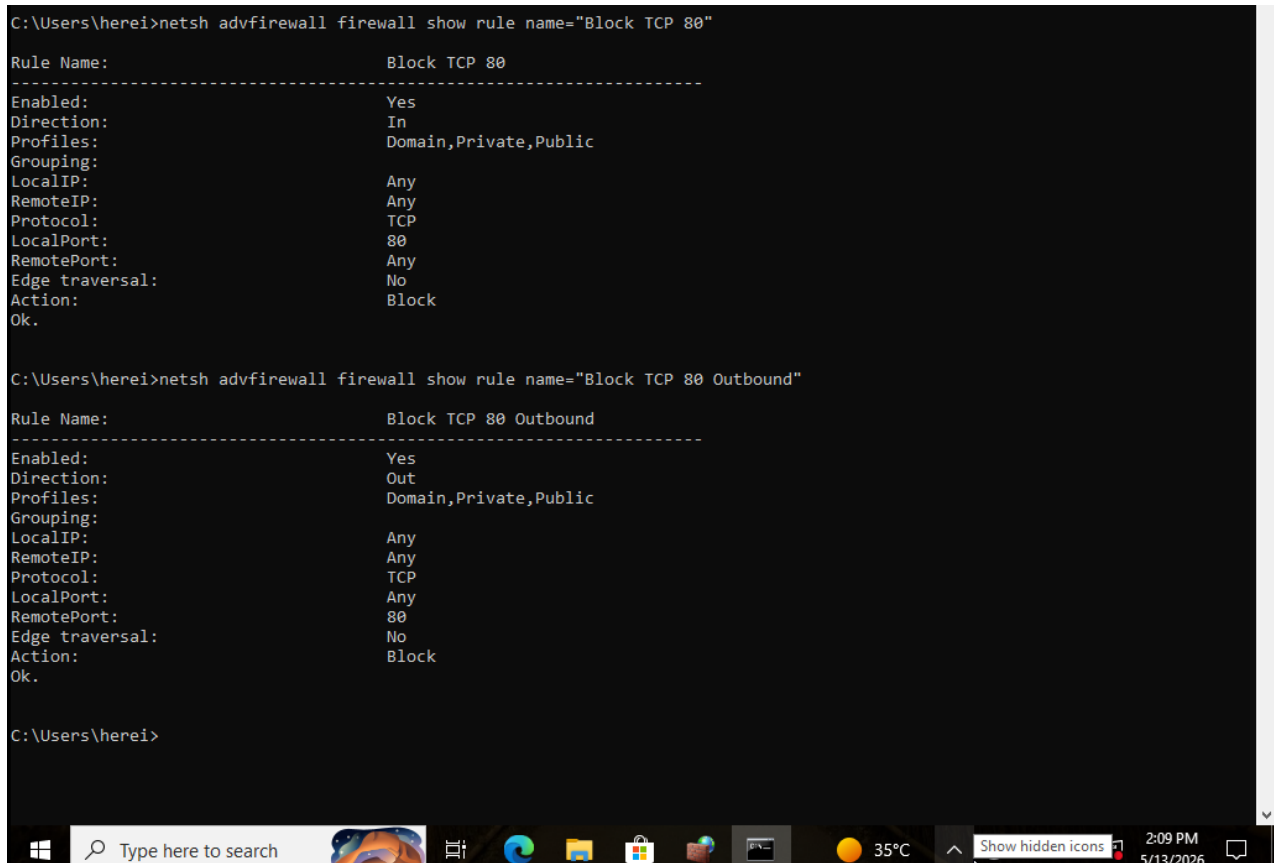
Date of Submission: May 13, 2026

Lab # 14**Task 1: Blocking Specific Ports (TCP and UDP) using GUI****1. Block TCP Port 80 (HTTP)****Inbound**

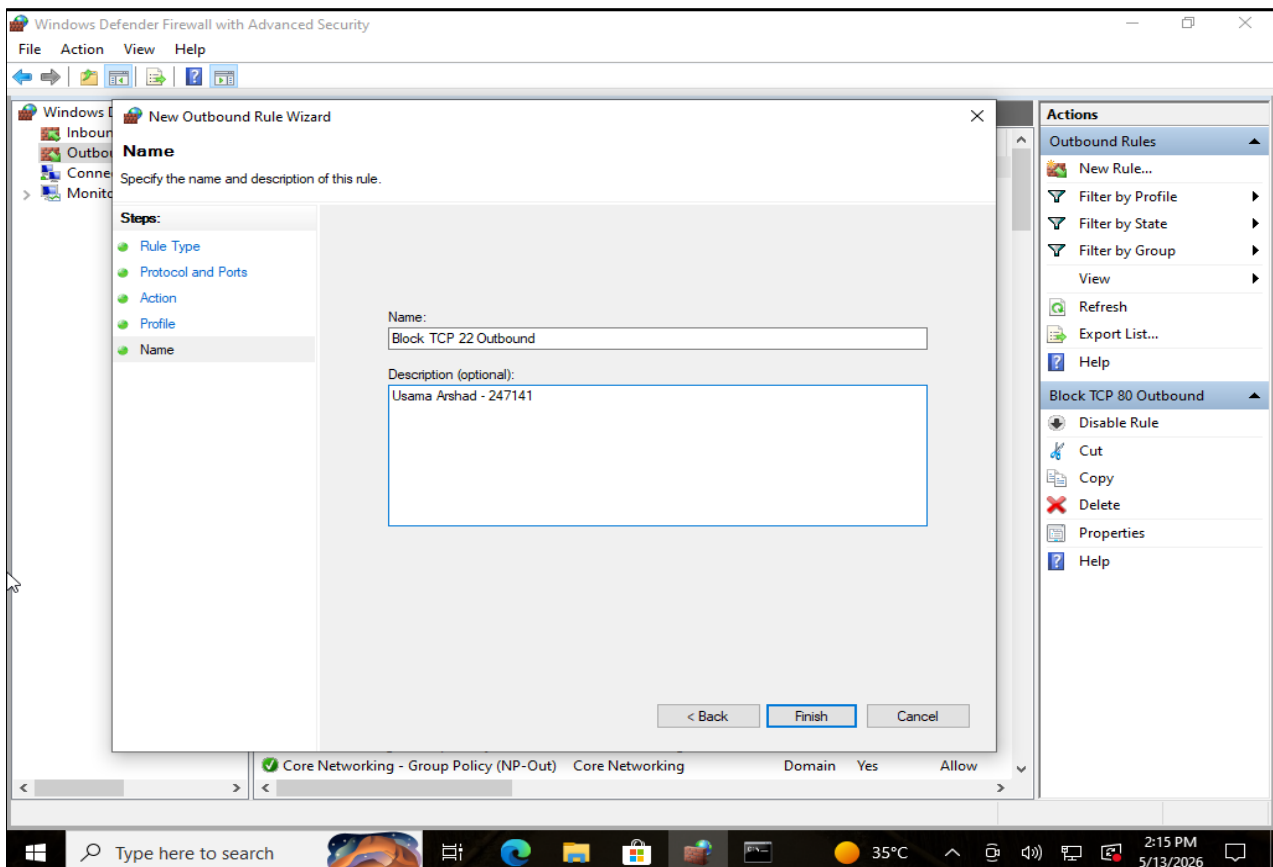
Outbound



Verification



Inbound



Verification

```
C:\Users\herei>netsh advfirewall firewall show rule name="Block TCP 22 Inbound"

Rule Name:                               Block TCP 22 Inbound
-----
Enabled:                                  Yes
Direction:                               In
Profiles:                                Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 TCP
LocalPort:                                22
RemotePort:                               Any
Edge traversal:                            No
Action:                                   Block
Ok.

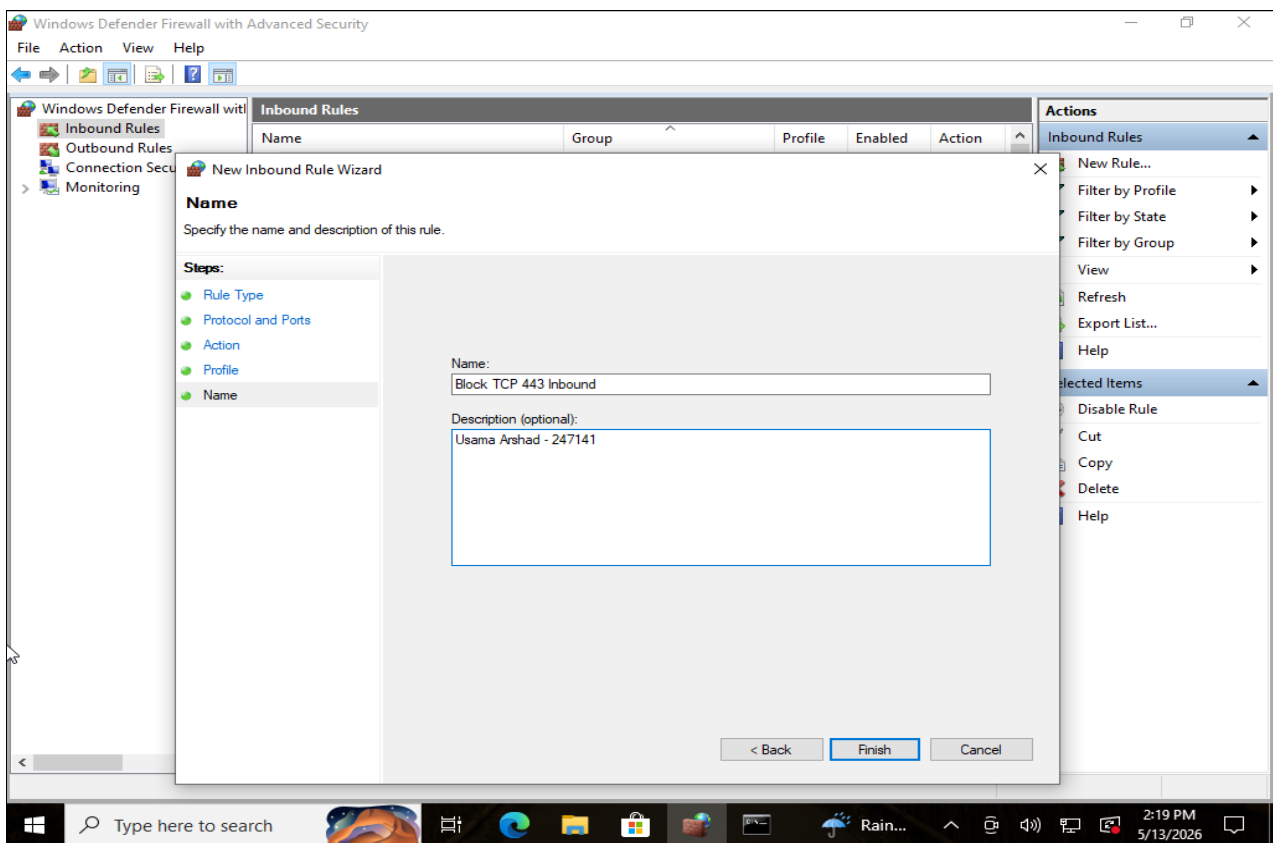
C:\Users\herei>netsh advfirewall firewall show rule name="Block TCP 22 Outbound"

Rule Name:                               Block TCP 22 Outbound
-----
Enabled:                                  Yes
Direction:                               Out
Profiles:                                Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 TCP
LocalPort:                                Any
RemotePort:                               22
Edge traversal:                            No
Action:                                   Block
Ok.

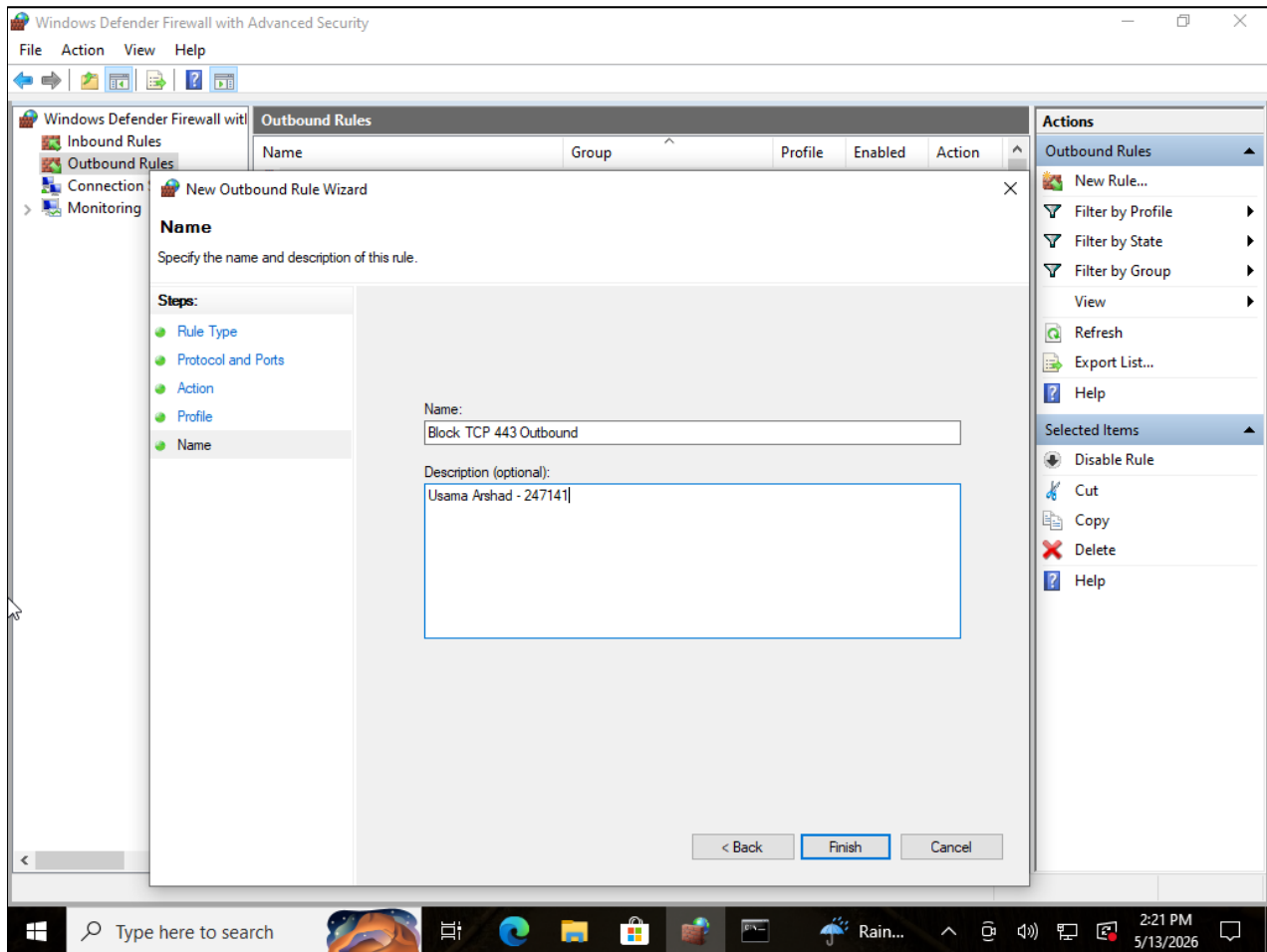
C:\Users\herei>
```

3. Block TCP Port 443 (HTTPS)

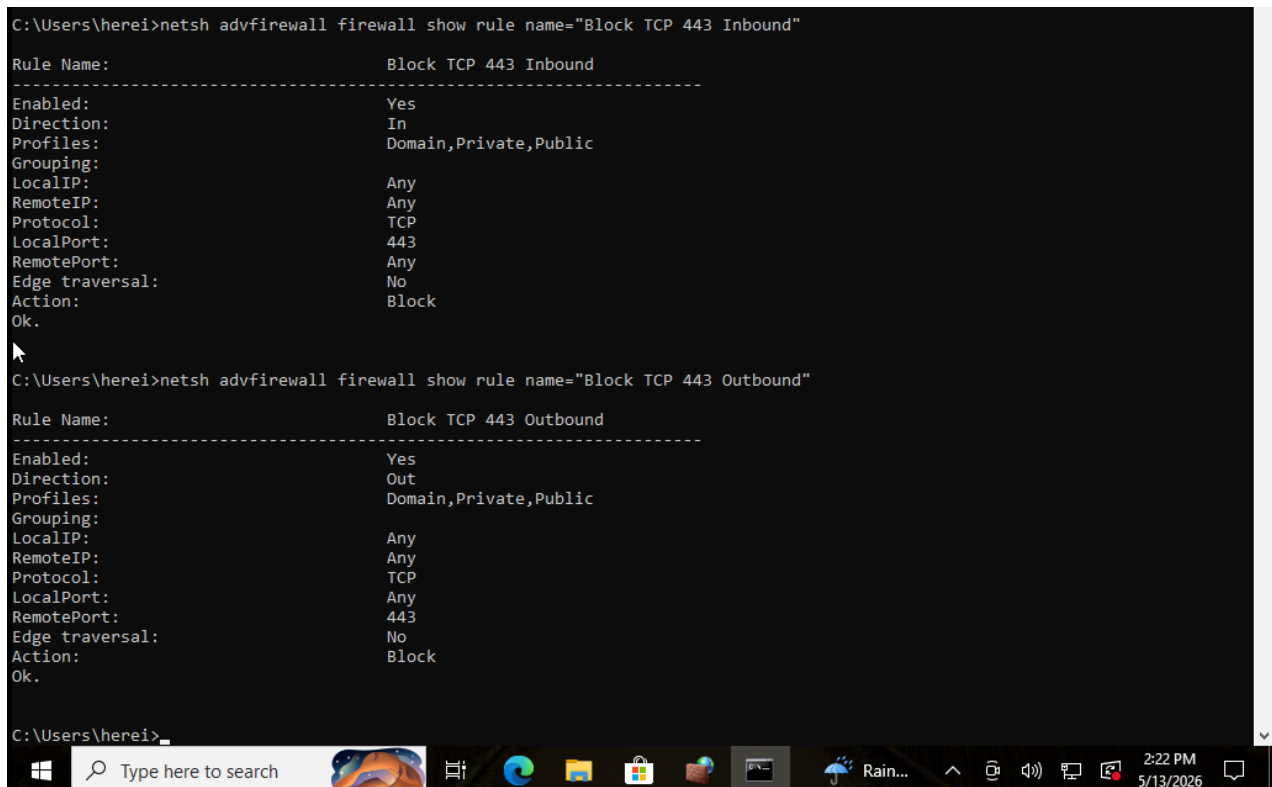
Inbound



Outbound

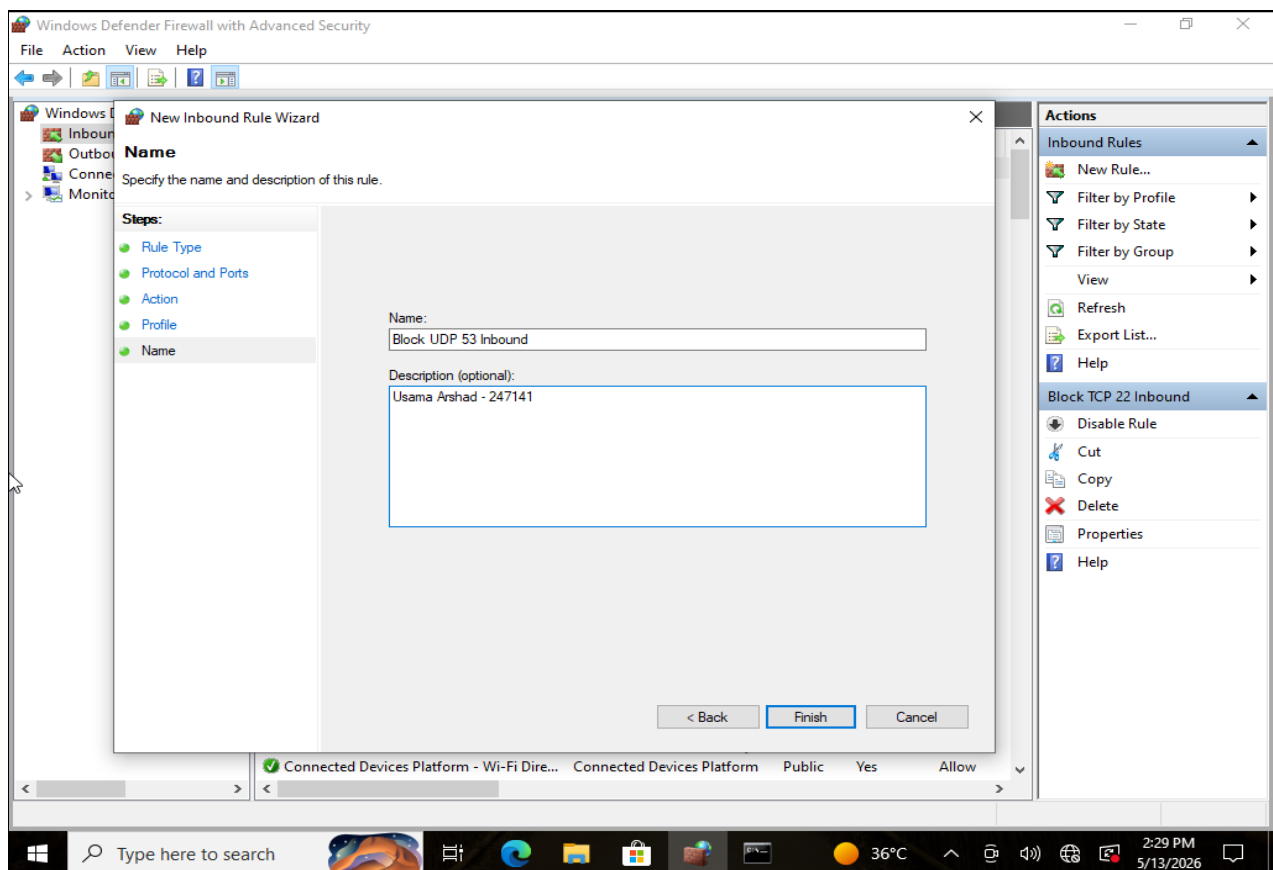


Verification

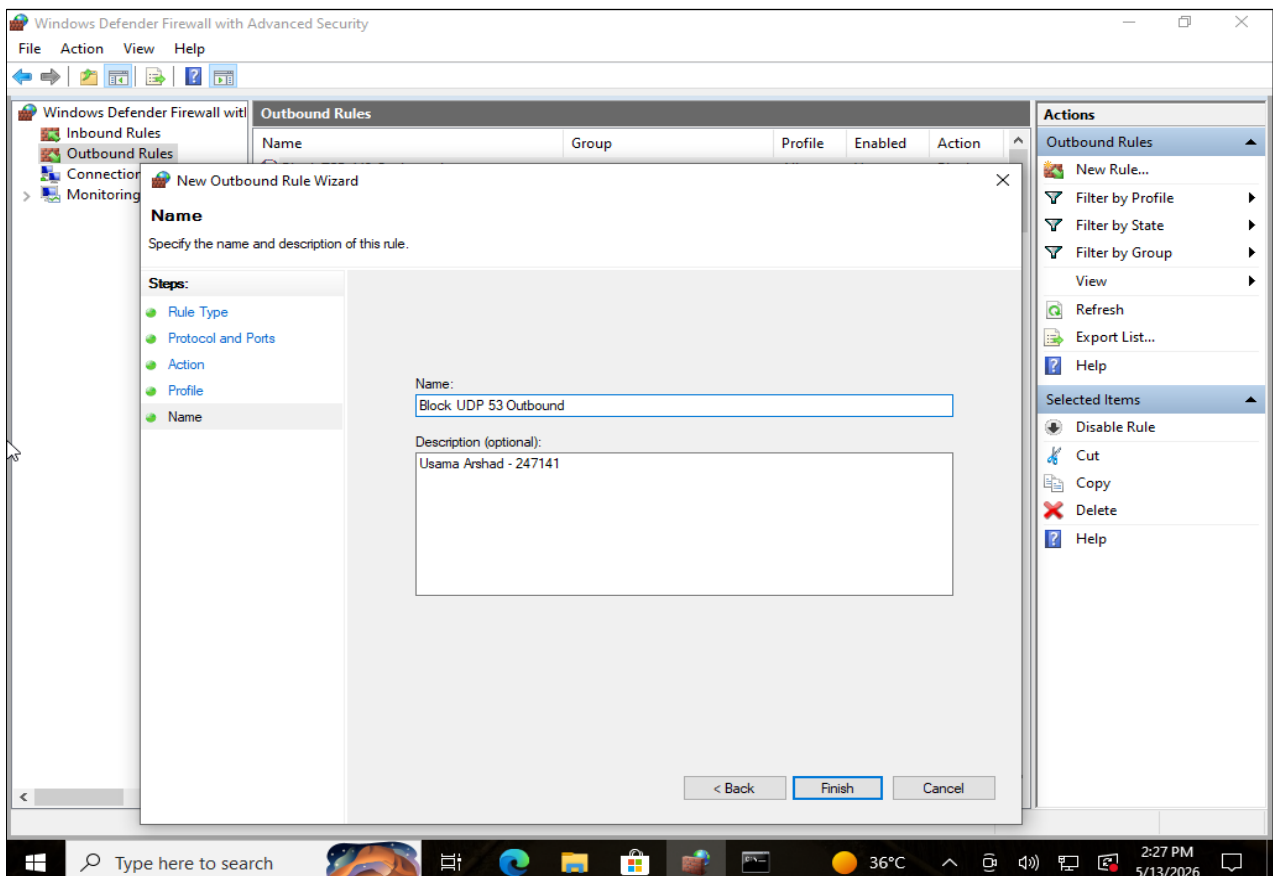


4. Block UDP Port 53 (DNS)

Inbound



Outbound



Verification

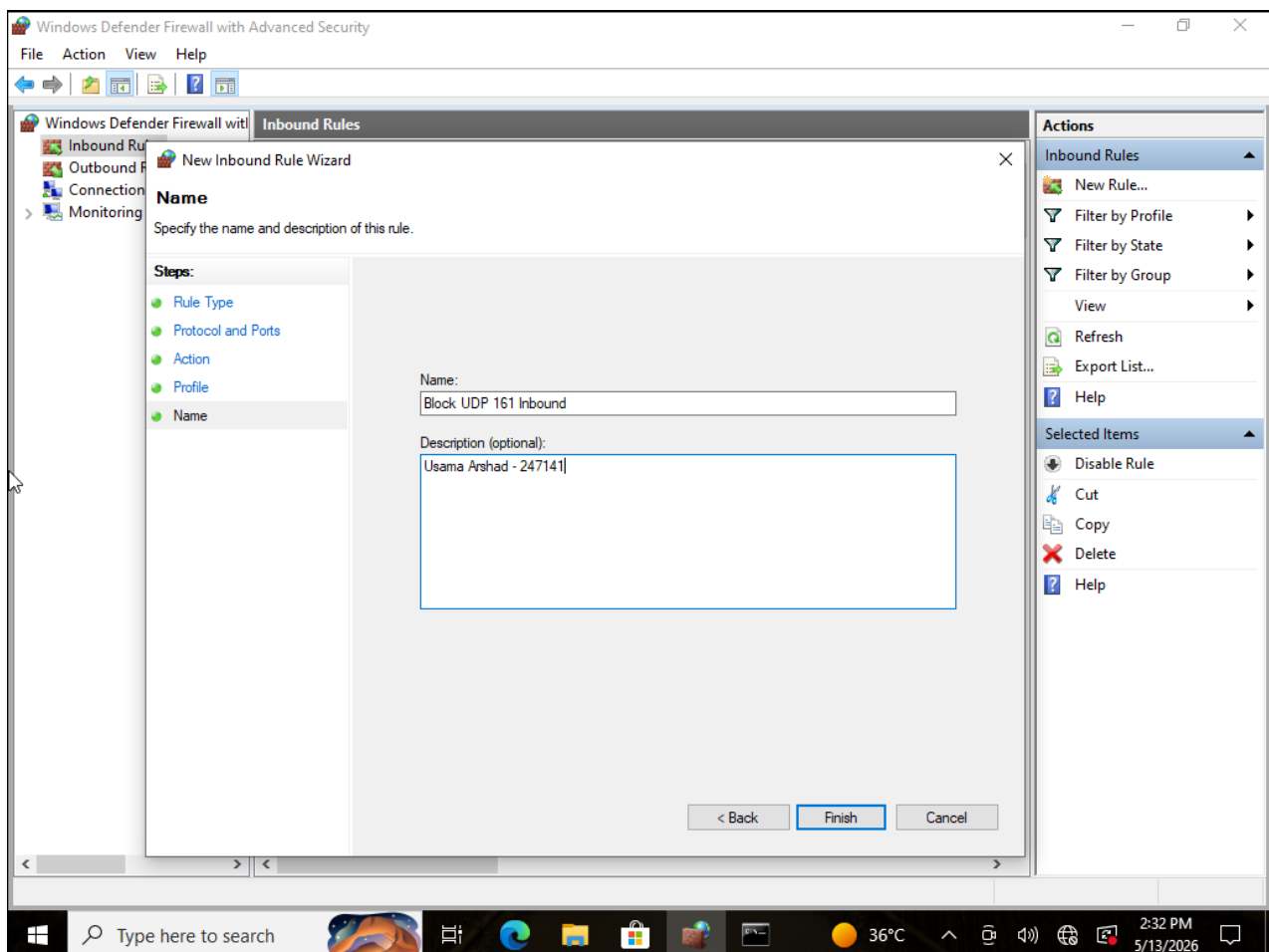
```
C:\Users\herei>netsh advfirewall firewall show rule name="Block UDP 53 Inbound"

Rule Name:                               Block UDP 53 Inbound
-----
Enabled:                                  Yes
Direction:                               In
Profiles:                                Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 TCP
LocalPort:                                53
RemotePort:                               Any
Edge traversal:                            No
Action:                                   Block
Ok.

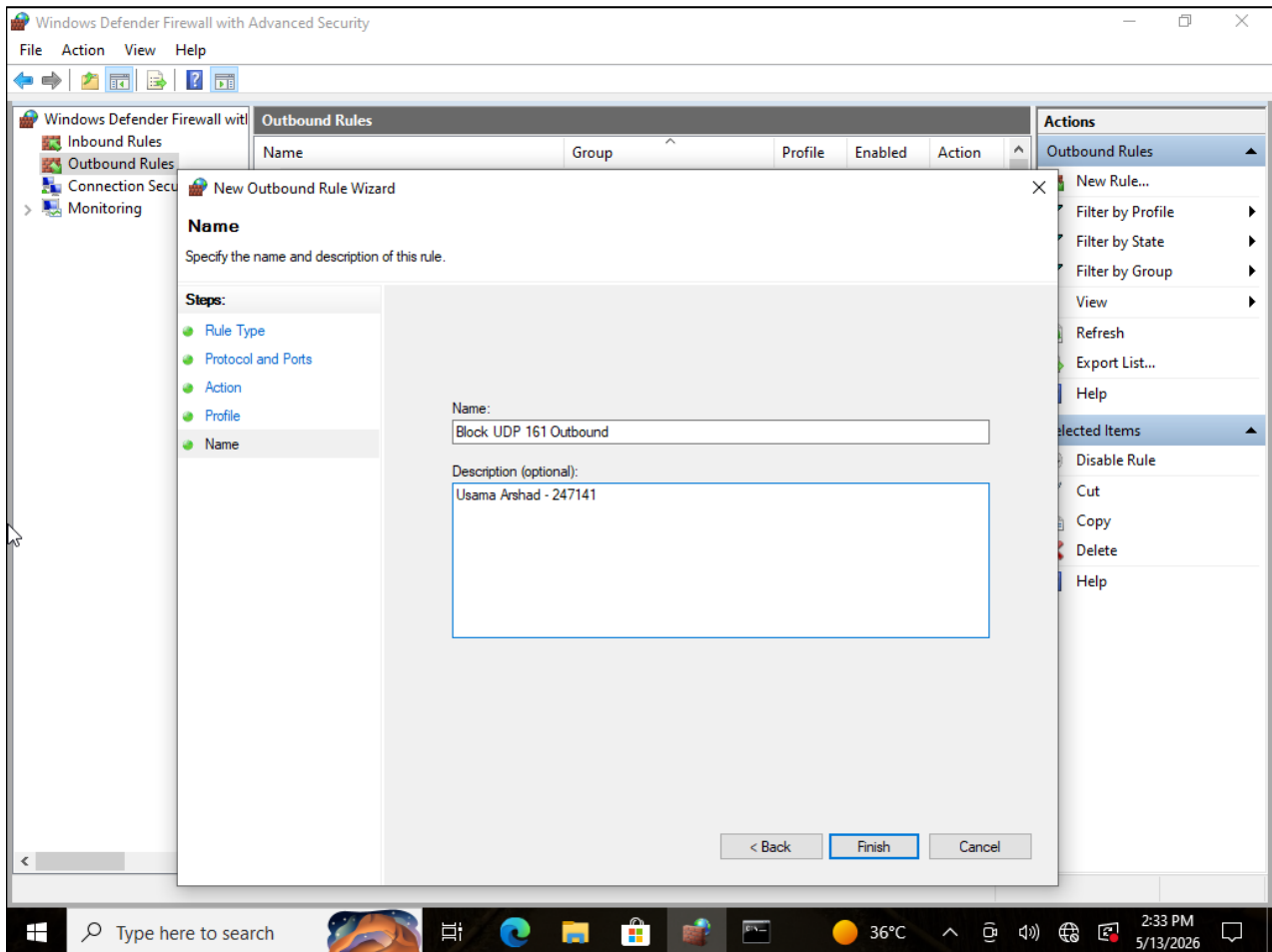
C:\Users\herei>netsh advfirewall firewall show rule name="Block UDP 53 Outbound"

Rule Name:                               Block UDP 53 Outbound
-----
Enabled:                                  Yes
Direction:                               Out
Profiles:                                Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 UDP
LocalPort:                                Any
RemotePort:                               53
Edge traversal:                            No
Action:                                   Block
Ok.

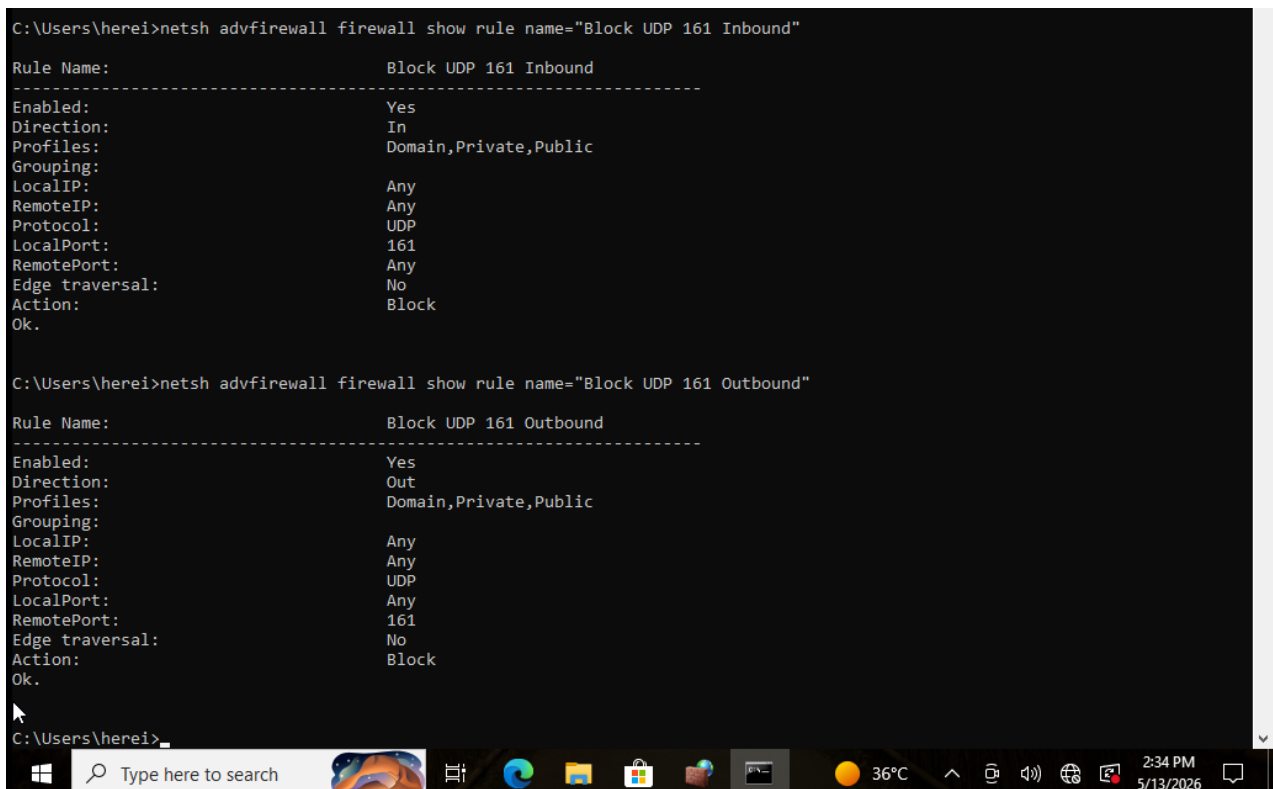
C:\Users\herei>
```

5. Block UDP Port 161 (SNMP)**Inbound**

Outbound

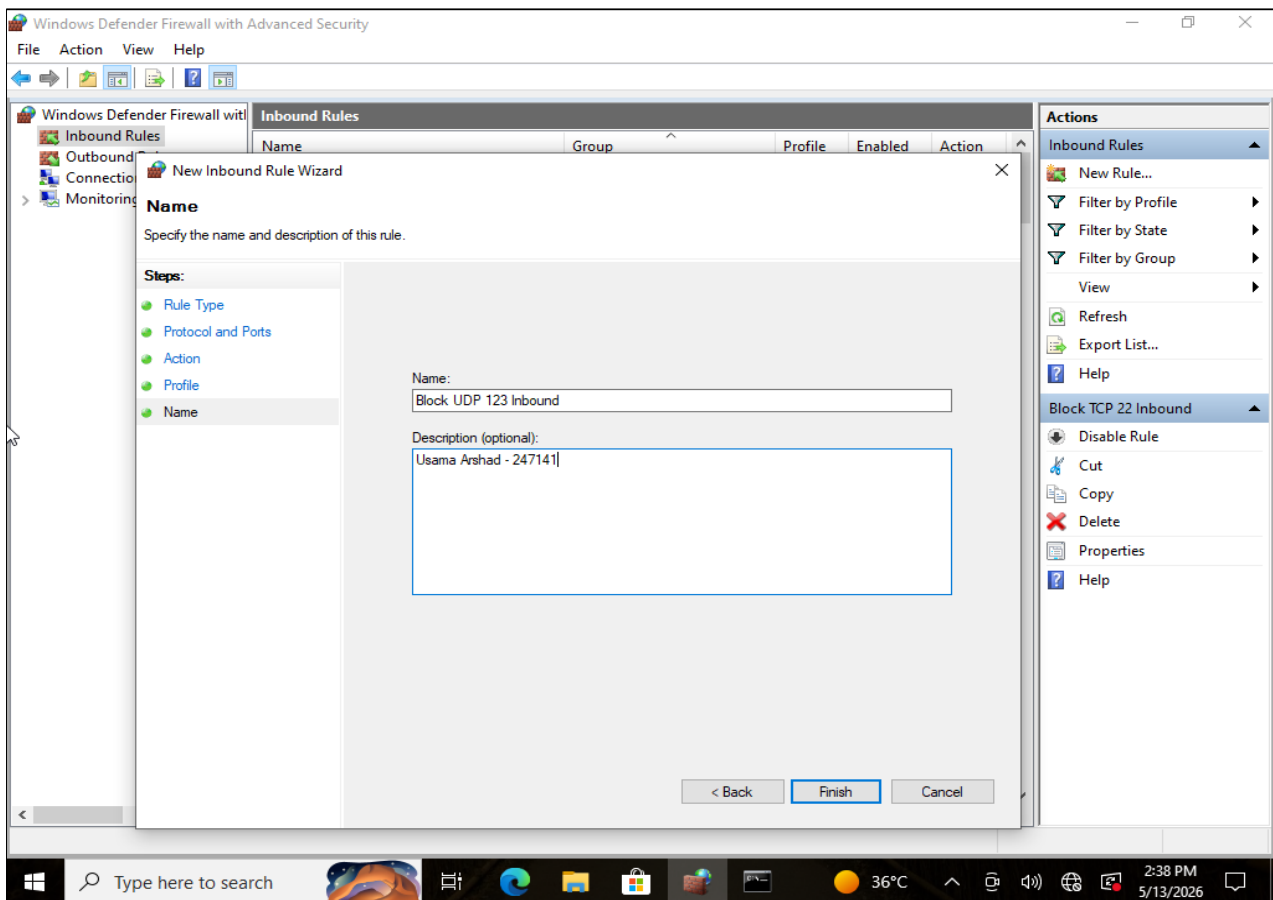


Verification

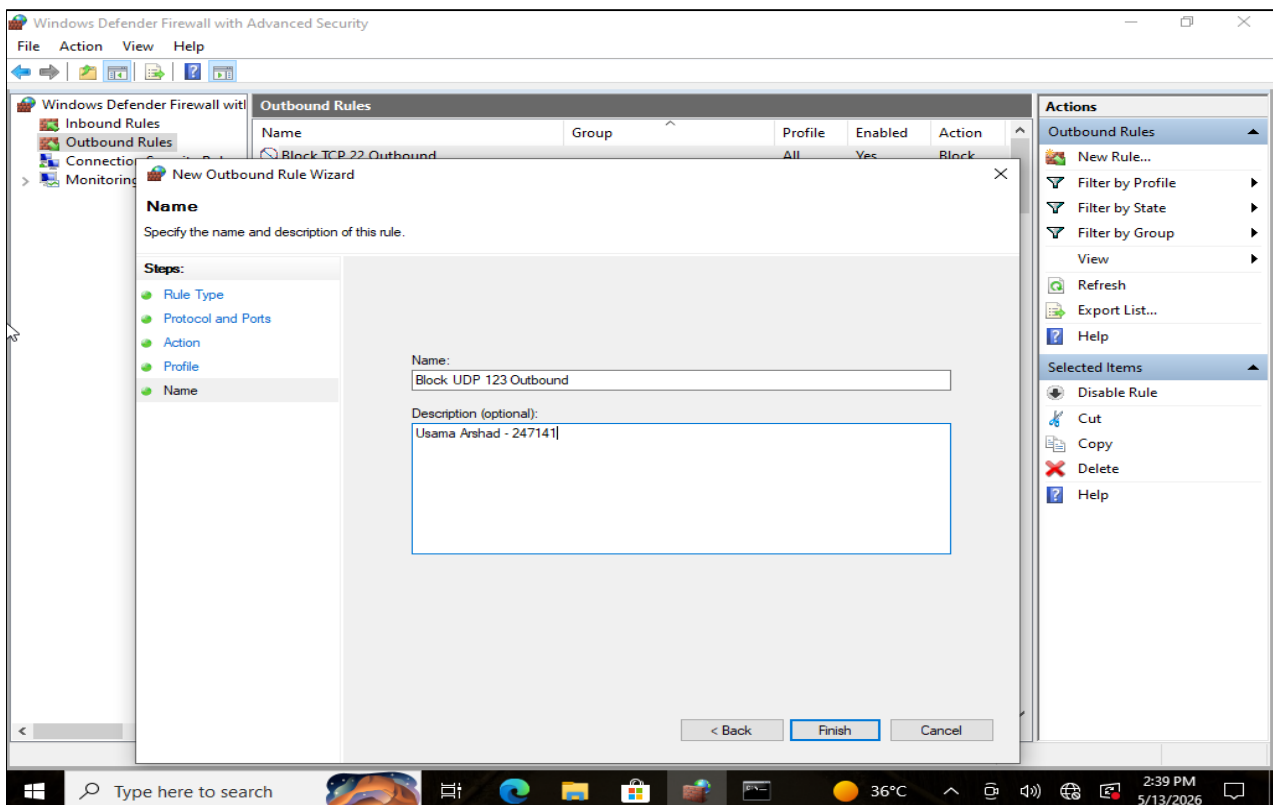


6. Block UDP Port 123 (NTP)

Inbound



Outbound



Verification

```

C:\Users\herei>netsh advfirewall firewall show rule name="Block UDP 123 Inbound"

Rule Name:                               Block UDP 123 Inbound
-----
Enabled:                                  Yes
Direction:                               In
Profiles:                                 Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 TCP
LocalPort:                                123
RemotePort:                               Any
Edge traversal:                            No
Action:                                   Block
Ok.

C:\Users\herei>netsh advfirewall firewall show rule name="Block UDP 123 Outbound"

Rule Name:                               Block UDP 123 Outbound
-----
Enabled:                                  Yes
Direction:                               Out
Profiles:                                 Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 TCP
LocalPort:                                Any
RemotePort:                               123
Edge traversal:                            No
Action:                                   Block
Ok.

```

7. Rules List

Inbound

Windows Defender Firewall with Advanced Security

File Action View Help

Windows Defender Firewall with Advanced Security

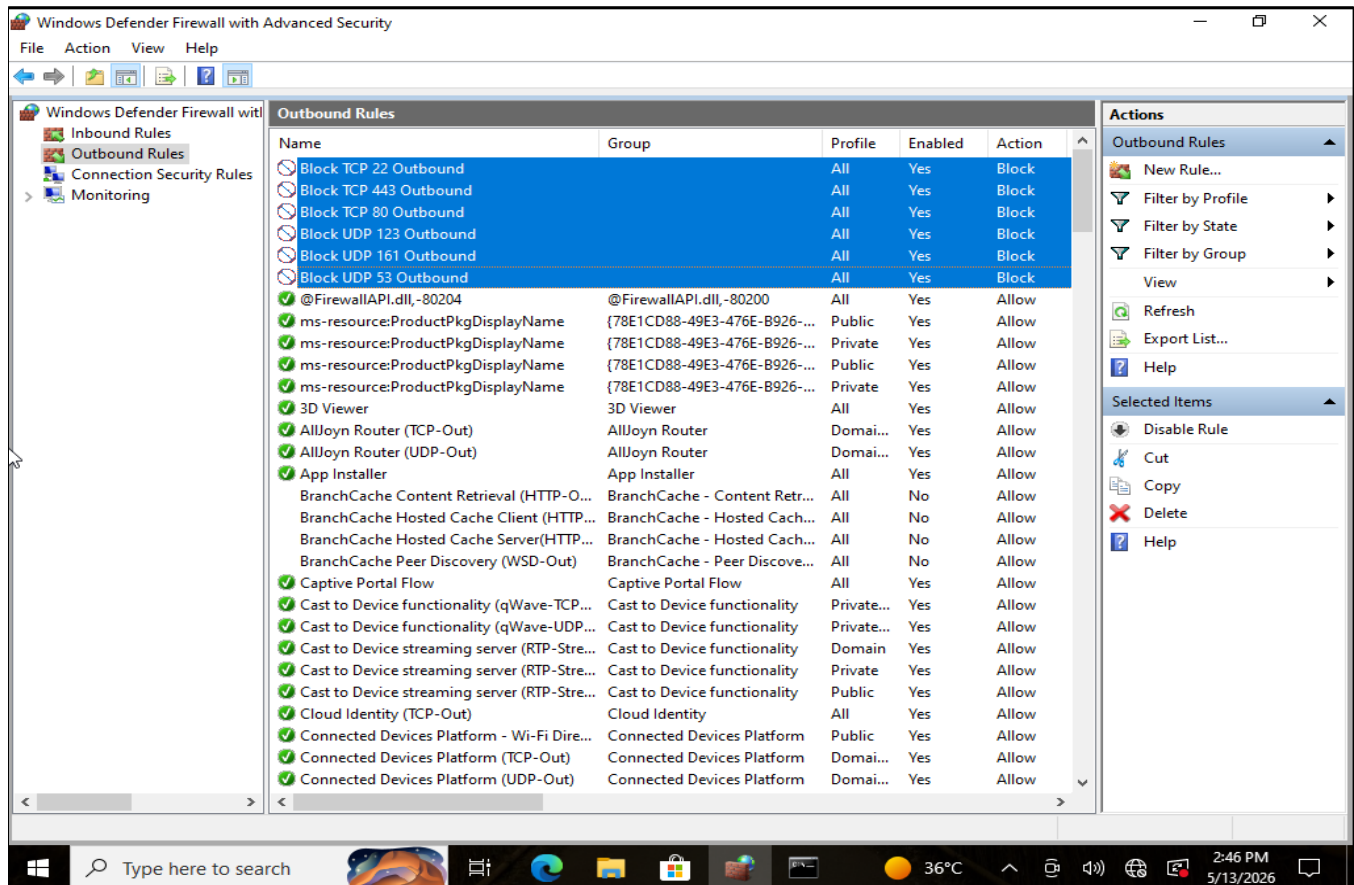
- Inbound Rules
- Outbound Rules
- Connection Security Rules
- Monitoring

Name	Group	Profile	Enabled	Action
Block TCP 22 Inbound		All	Yes	Block
Block TCP 443 Inbound		All	Yes	Block
Block TCP 80		All	Yes	Block
Block UDP 123 Inbound		All	Yes	Block
Block UDP 161 Inbound		All	Yes	Block
Block UDP 53 Inbound		All	Yes	Block
@FirewallAPI.dll, -80201	@FirewallAPI.dll, -80200	All	Yes	Allow
@FirewallAPI.dll, -80206	@FirewallAPI.dll, -80200	All	Yes	Allow
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...}	Public	Yes	Allow
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...}	Private	Yes	Allow
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...}	Public	Yes	Allow
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...}	Private	Yes	Allow
AllJoyn Router (TCP-In)	AllJoyn Router	Domai...	Yes	Allow
AllJoyn Router (UDP-In)	AllJoyn Router	Domai...	Yes	Allow
App Installer	App Installer	Domai...	Yes	Allow
BranchCache Content Retrieval (HTTP-In)	BranchCache - Content Retr...	All	No	Allow
BranchCache Hosted Cache Server (HTTP...	BranchCache - Hosted Cach...	All	No	Allow
BranchCache Peer Discovery (WSD-In)	BranchCache - Peer Discove...	All	No	Allow
Cast to Device functionality (qWave-TCP...	Cast to Device functionality	Private...	Yes	Allow
Cast to Device functionality (qWave-UDP...	Cast to Device functionality	Private...	Yes	Allow
Cast to Device SSDP Discovery (UDP-In)	Cast to Device functionality	Public	Yes	Allow
Cast to Device streaming server (HTTP-St...	Cast to Device functionality	Public	Yes	Allow
Cast to Device streaming server (HTTP-St...	Cast to Device functionality	Private	Yes	Allow
Cast to Device streaming server (HTTP-St...	Cast to Device functionality	Domain	Yes	Allow
Cast to Device streaming server (RTCP-Str...	Cast to Device functionality	Public	Yes	Allow
Cast to Device streaming server (RTCP-Str...	Cast to Device functionality	Private	Yes	Allow
Cast to Device streaming server (RTCP-Str...	Cast to Device functionality	Domain	Yes	Allow
Cast to Device streaming server (RTSP-Str...	Cast to Device functionality	Public	Yes	Allow
Cast to Device streaming server (RTSP-Str...	Cast to Device functionality	Private	Yes	Allow

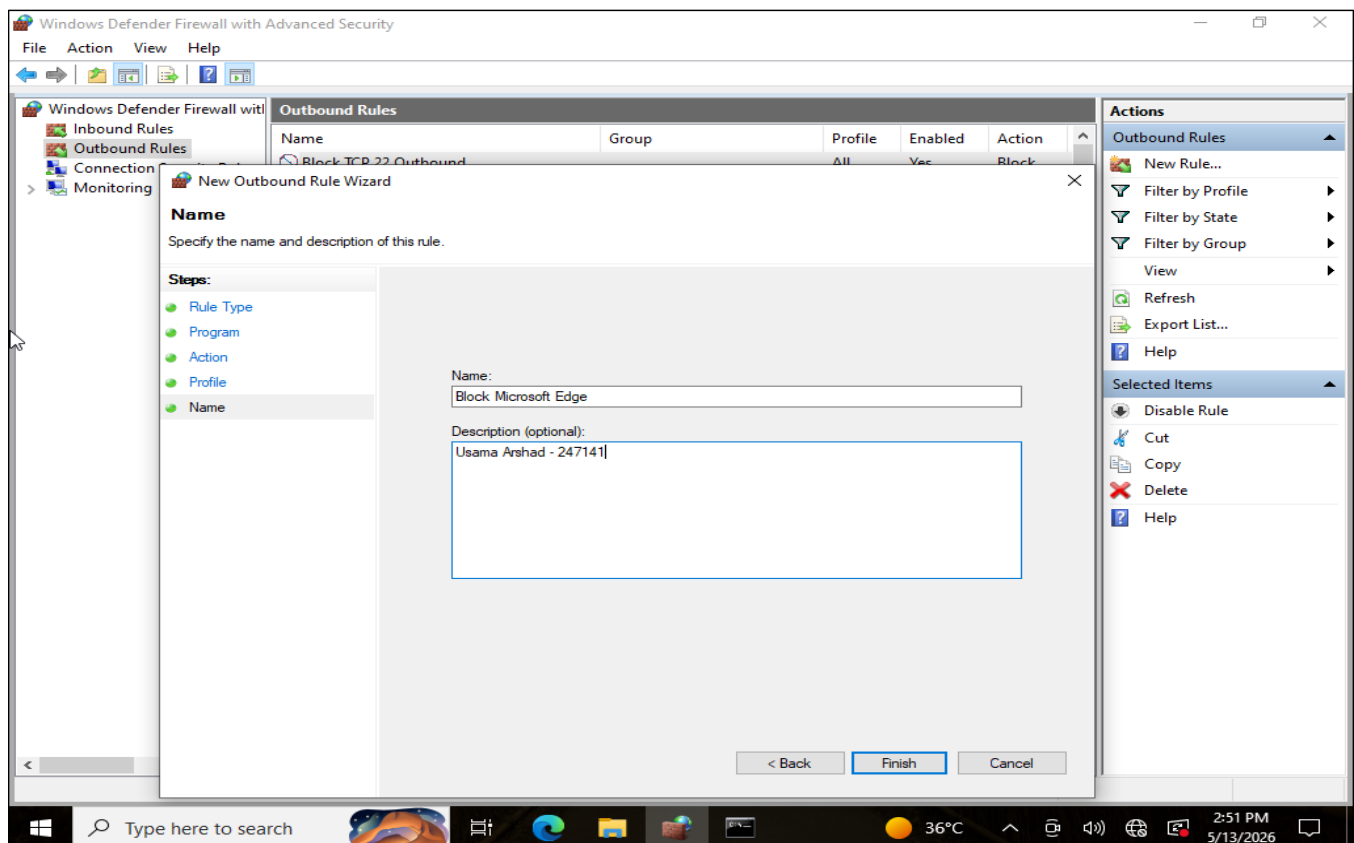
Actions

- Inbound Rules
- New Rule...
- Filter by Profile
- Filter by State
- Filter by Group
- View
- Refresh
- Export List...
- Help
- Selected Items
- Disable Rule
- Cut
- Copy
- Delete
- Help

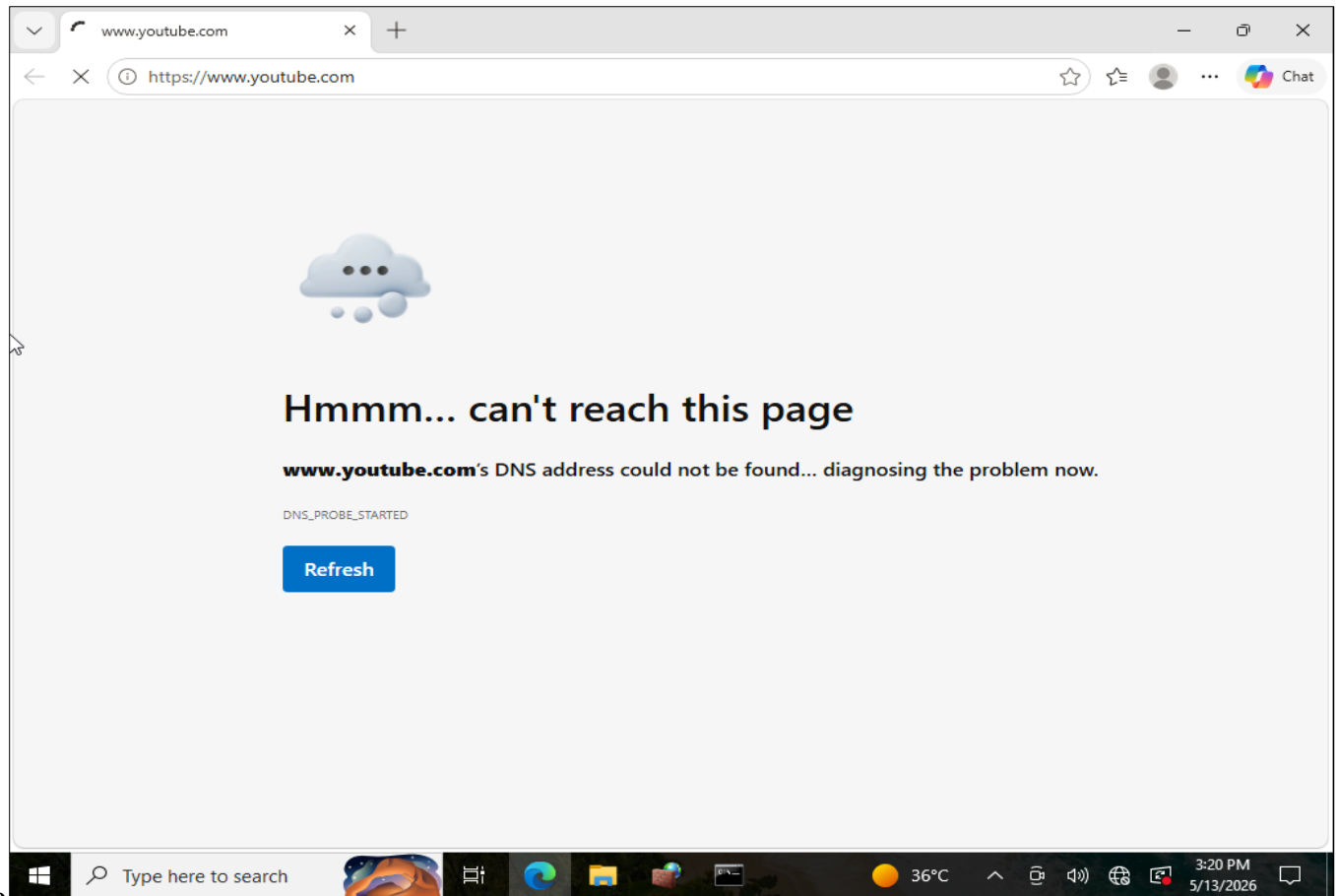
Outbound



Task 2: Blocking Microsoft Edge Access using the GUI



Verification



Importance of Firewall Rules in an Organization

Blocking Port 80 (HTTP) and Port 443 (HTTPS): Blocking these ports on devices that are not authorized helps stop unwanted access to the internet. In places like military and financial organizations, controlling web traffic is important for protecting data and preventing unauthorized communication.

Blocking Port 22 (SSH): SSH (Secure Shell) is used for remote access to systems. Blocking this port can stop attackers inside the network from moving to other machines or connecting to outside servers.

Blocking Port 53 (DNS): The Domain Name System (DNS) is often misused by malware for data theft and to control systems through DNS tunneling. Blocking unauthorized DNS traffic makes sure that all DNS requests go to a monitored server, which helps identify possible attacks.

Blocking Port 161 (SNMP): SNMP (Simple Network Management Protocol) is used to manage network devices. If this port is open, attackers can gather detailed information about the network from routers, switches, and servers. Blocking this port reduces chances for network spying.

Blocking Port 123 (NTP): NTP (Network Time Protocol) is used to synchronize system time. Open NTP can be used by attackers for amplification DDoS (Distributed Denial of Service) attacks. Blocking unauthorized NTP traffic protects both the organization and other possible targets.

Blocking a Specific Application (Microsoft Edge)

Administrators may need to block certain browsers or applications for different reasons. This helps ensure that only approved and secure browsers are used, preventing employees from bypassing web filters or using unauthorized applications. Blocking at the application level is usually more effective than just blocking ports because applications can run on multiple ports. By focusing on specific programs, security measures are stronger.

- End of Lab Task -